

Summary of 1709.04891 §2 & 3

- particles are irreducible ^{unitary} rep of Poincare group
- a particle is labeled by its momentum and transforms under some rep of the little group.
- awkwardness in describing particles by fields: fields transform as Lorentz tensors or spinors, while particles transform under the little group.
- Feynman diagrams have the wrong transformation properties to be called scattering amplitudes - need to introduce polarization vectors/spinors that satisfy some constraints to get a proper scattering amplitude

By spin-1 massless polarization vectors do not transform genuinely as Lorentz vectors: $(\Lambda p) = p$,
 $(\Lambda \epsilon_{\pm}) = e^{\pm i\theta} \epsilon_{\pm} + \alpha(p) p$ - that is only the gauge equivalence class $\{\epsilon_{\pm} | \epsilon_{\pm} + \alpha p\}$ is invariant under Lorentz transformations

- spin helicity variables transform nicely under both little group & Lorentz group transformations as well as represent the constraint on shell momenta $P_{\mu} \dot{x}^{\mu}$ by unconstrained $\lambda \dot{\alpha} \dot{\bar{\alpha}}$. - Amplitudes are functions of $\lambda \dot{\alpha} \dot{\bar{\alpha}}$

- This chapter teaches us how scattering amplitudes factorize on a pole (amp only has simple poles in s, t, u)

$$\lim_{\substack{x \rightarrow 0 \\ x \in \{s, t, u\}}} x \mathcal{M}(s, t, u) = \mathcal{M}_L(s, t, u) \mathcal{M}_R(s, t, u)$$

- often residues in one channel have poles in another and the consistent factorization of amplitudes place strict constraints on it.

Ex1 there does not exist a consistent 4pt amplitude of massless particles except for those of spin $0, \frac{1}{2}, 1, \frac{3}{2}, 2$

Ex2 self interacting spin 1 particles need a color structure to be consistent

Ex3 self-interacting spin 2 particles do not have a color structure

Ex4 cannot have a consistent theory of massless charged particles with spin $\geq \frac{3}{2}$

Ex5 spin > 2 cannot consistently couple to gravity

Ex6 no coupling b/w 3 particles such that $h_1 + h_2 + h_3 = 0$.

3.7

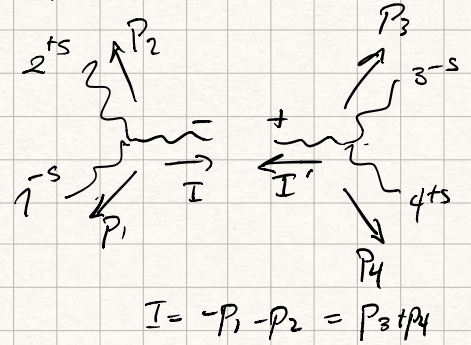
Self-interacting massless particle of spin S : $A_3 [1^{-S} 2^S 3^{+S}] = \frac{\langle 12 \rangle^{SS}}{\langle 13 \rangle^S \langle 23 \rangle^S}$

look at S -channel pole: $(p_1 + p_2)^2 = \langle 12 \rangle [12] = \langle 34 \rangle [34] = (p_3 + p_4)^2 \rightarrow 0$

$A_4 [1^{-S} 2^{+S} 3^{-S} 4^{+S}] = \mathcal{F} \langle 13 \rangle^2 [24]^2$ by little group scaling

choose $[12] \rightarrow 0$ and $\langle 34 \rangle \rightarrow 0$

$$[A_4] = 0 = [\mathcal{F}] + 4 \Rightarrow [\mathcal{F}] = -4$$



S-channel

$$\begin{aligned} \lim_{S \rightarrow 0} S A_4 &= A_3 [1^{-S} 2^{+S} I^{-S}] A_3 [3^{-S} 4^{+S} I'^{+S}] \\ &= \left(\frac{\langle 1 I \rangle^3}{\langle 12 \rangle \langle I2 \rangle} \right)^S \left(\frac{[4 I']^3}{[I3] [43]} \right)^S \\ &= \left(\frac{(\langle 12 \rangle [24])^3}{\langle 12 \rangle \langle I2 \rangle [I3] [43]} \right)^S \\ &= \left(\frac{(-\langle 13 \rangle [34])^2 \langle I2 \rangle [24]}{\langle I2 \rangle \langle I2 \rangle [I3] [43]} \right)^S \\ &= (\langle 13 \rangle^2 [24]^2)^S \left(\frac{[34]^2}{\langle I2 \rangle [I3] [43] [24]} \right)^S \\ &= (\langle 13 \rangle^2 [24]^2)^S \left(\frac{[34]^2}{\langle 24 \rangle [43]^2 [24]} \right)^S \\ &= \left(\frac{\langle 13 \rangle^2 [24]^2}{u} \right)^S \end{aligned}$$

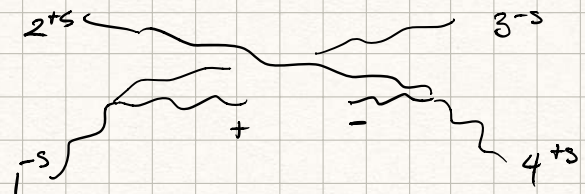
① $1) (1+2)(2+3)(3+4)(4+1) = 0$
 $\langle 12 \rangle [24] + \langle 13 \rangle [34] = 0$
 $\langle 12 \rangle [24] = -\langle 13 \rangle [34]$

② $3) [3+4][4+I][I] = 0$
 $\langle 24 \rangle [43] + \langle 2I \rangle [I3] = 0$
 $\langle I2 \rangle [I3] = +\langle 24 \rangle [43]$

③ $u = \langle 13 \rangle [13] = \langle 24 \rangle [24]$

U-channel

$$\begin{aligned} \lim_{u \rightarrow 0} u A_4 [1^{-S} 2^{+S} 3^{-S} 4^{+S}] &= A_3 [1^{-S} 3^{-S} I^{+S}] A_3 [2^{+S} 4^{+S} I^{-S}] \\ &= \left(\frac{\langle 13 \rangle^3}{\langle 1I \rangle \langle 3I \rangle} \right)^S \left(\frac{[24]^3}{[2I] [4I]} \right)^S \\ &= (\langle 13 \rangle^2 [24]^2)^S \left(\frac{\langle 13 \rangle [24]}{\langle 1I \rangle \langle 3I \rangle [2I] [4I]} \right)^S \\ &= (\langle 13 \rangle^2 [24]^2)^S \left(\frac{\langle 13 \rangle [24]}{\langle 13 \rangle [32] \langle 32 \rangle [24]} \right)^S \\ &= \left(\frac{\langle 13 \rangle [24]^2}{t} \right)^S \end{aligned}$$

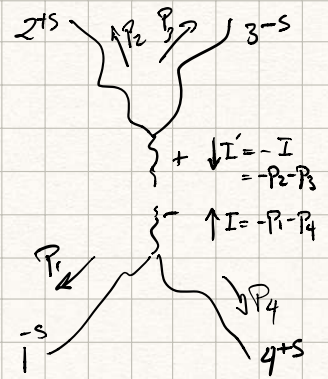


1) $[1+3][3+I][I] = 0$
2) $[2+4][4+I][I] = 0$

$\langle 1I \rangle [2I] = \langle 13 \rangle [32]$
 $\langle 3I \rangle [4I] = \langle 32 \rangle [24]$
 $t = \langle 14 \rangle [14] = \langle 23 \rangle [23]$

the t channel pole: $t = (p_1 + p_4)^2 = \langle 14 \rangle [14] = \langle 23 \rangle [23] = (p_2 + p_3)^2 \rightarrow 0$

$$\begin{aligned}
 \lim_{t \rightarrow 0} t A_4 [1^{-s} 2^{+s} 3^{-s} 4^{+s}] \\
 &= A_8 [1^{-s} I^{+s} 4^{+s}] A_3 [2^{+s} I^{+s} 3^{-s}] \\
 &= \left(\frac{\langle 1I \rangle^3 [2I]^3}{\langle 14 \rangle \langle I4 \rangle [23] [I3]} \right)^s \\
 &= \left(\frac{-\langle 13 \rangle^2 [32]^2 \langle 4I \rangle [42]}{\langle 4I \rangle [23] \langle I4 \rangle [I3]} \right)^s \\
 &= \left(\frac{+\langle 13 \rangle^2 [32] [24]}{\langle I4 \rangle [I3]} \right)^s \\
 &= \left(\frac{\langle 13 \rangle^2 [24]^2}{s} \right)^s
 \end{aligned}$$



$$\begin{aligned}
 2) [2 + 3] [3 + I] [I = 0] \\
 1) [1 + 4] [4 + I] [I = 0]
 \end{aligned}$$

$$R_s = \left(\frac{\langle 13 \rangle^2 [24]^2}{s} \right)^s \quad R_u = \left(\frac{\langle 13 \rangle [24]^2}{t} \right)^s \quad R_t = \left(\frac{\langle 13 \rangle [24]^2}{s} \right)^s$$

3.10

We know that

$$\begin{aligned}\lim_{s \rightarrow 0} s \mathcal{F} &< 13 \rangle^2 [24]^2 = \frac{\langle 13 \rangle^2 [24]^2}{u} = R_s \\ \lim_{t \rightarrow 0} t \mathcal{F} &< 13 \rangle^2 [24]^2 = \frac{\langle 13 \rangle^2 [24]^2}{s} = R_t \\ \lim_{u \rightarrow 0} u \mathcal{F} &< 13 \rangle^2 [24]^2 = \frac{\langle 13 \rangle^2 [24]^2}{t} = R_u\end{aligned}$$

Residues of one channel have poles in the other channels.

$$\Rightarrow \mathcal{F} = \left(\frac{A}{st} + \frac{B}{tu} + \frac{C}{us} \right)$$

as $s \rightarrow 0$, $t \rightarrow -u \Rightarrow$ the residue in the s channel is:

$$\begin{aligned}\lim_{s \rightarrow 0} s \left(\frac{A}{us} + \frac{B}{tu} + \frac{C}{us} \right) &< 13 \rangle^2 [24]^2 \\ &= \left(\frac{C}{u} - \frac{A}{u} \right) < 13 \rangle^2 [24]^2 \\ &= \frac{\langle 13 \rangle^2 [24]^2}{u} \Rightarrow C - A = 1\end{aligned}$$

as $t \rightarrow 0$, $u \rightarrow -s$ and:

$$\begin{aligned}\lim_{t \rightarrow 0} A_t &= \left(\frac{A}{s} - \frac{B}{s} \right) < 13 \rangle^2 [24]^2 \\ &= \frac{\langle 13 \rangle^2 [24]^2}{s} \Rightarrow A - B = 1\end{aligned}$$

as $u \rightarrow 0$, $s \rightarrow -t$, and:

$$\begin{aligned}\lim_{u \rightarrow 0} A_u &= \left(\frac{B}{t} - \frac{C}{t} \right) < 13 \rangle^2 [24]^2 \\ &= \frac{\langle 13 \rangle^2 [24]^2}{t} \Rightarrow B - C = 1\end{aligned}$$

No solution

need more than one flavour.

Self interaction: $A_3 [1^{-s} 2^{+s} 3^{-s}] \rightarrow f^{a_1 a_2 a_3} \left(\frac{\langle 13 \rangle^2}{\langle 12 \rangle \langle 23 \rangle} \right)^s$.

We know that f must be totally asym b/c the brackets are totally asym and this is an amp for bosons.

$$\begin{aligned}\Rightarrow R_s &\rightarrow f^{a_1 a_2 b} f^{a_3 a_4 b} R_s \\ R_t &\rightarrow R_t \\ R_u &\rightarrow R_u\end{aligned}$$

Diagrammatic representation of the above equations using Feynman diagrams with external legs labeled with spin and indices.

Ansatz: $\mathcal{F} = \left(\frac{A^{a_1 a_2 a_3 a_4}}{st} + \frac{B^{a_1 a_2 a_3 a_4}}{tu} + \frac{C^{a_1 a_2 a_3 a_4}}{us} \right)$

from $s \rightarrow 0$: $C^{a_1 a_2 a_3 a_4} - A^{a_1 a_2 a_3 a_4} = f^{a_1 a_2 b} f^{b a_3 a_4}$
 " $t \rightarrow 0$: $A^{a_1 a_2 a_3 a_4} - B^{a_1 a_2 a_3 a_4} = f^{a_1 a_4 b} f^{b a_3 a_2} = f^{a_2 a_3 b} f^{b a_4 a_1}$
 " $u \rightarrow 0$: $B^{a_1 a_2 a_3 a_4} - C^{a_1 a_2 a_3 a_4} = f^{a_1 b a_3} f^{a_2 b a_4} = f^{a_1 a_3 b} f^{b a_2 a_4}$

3.12

$S=2$: would think that there is a $1/u^2$ pole in the s -channel residue.
but there is only an $S \rightarrow \infty$ We could just as well write
 $1/u^2 = -1/tu$

$\Rightarrow$ unique 4pt amplitude: $-\frac{\langle 13 \rangle^4 [24]^4}{stu}$

$$A_3 [1^{-2} 2^{-2} 3^{+2}] = \left(\frac{\langle 12 \rangle^3}{\langle 13 \rangle \langle 23 \rangle} \right)^2$$

We consider the 4pt amp: $A_4 [1^{-2} 2^{+2} 3^{-2} 4^{+2}] = \mathcal{F} (\langle 13 \rangle [24])^4$

$$R_s = \left(\frac{\langle 13 \rangle^2 [24]^2}{u} \right)^2, \quad R_t = \left(\frac{\langle 13 \rangle^2 [24]^2}{s} \right)^2, \quad R_u = \left(\frac{\langle 13 \rangle [24]^2}{t} \right)^2$$

$$\Rightarrow \mathcal{F} = \left(\frac{A}{su^2} + \frac{B}{ts^2} + \frac{C}{ut^2} \right)$$

but using $s+t=0$, in the $s \rightarrow 0$, $t \rightarrow 0$, and $u \rightarrow 0$ limits
we can write

$$\begin{aligned} \mathcal{F} &= -\frac{(A+B+C)}{stu} \\ &= -\frac{1}{stu} \end{aligned}$$

and

$$A_4 = -\frac{\langle 34 \rangle^4 [24]^2}{stu}$$

3.19 3 pt. amp for gravity and 2 spin-5 particles

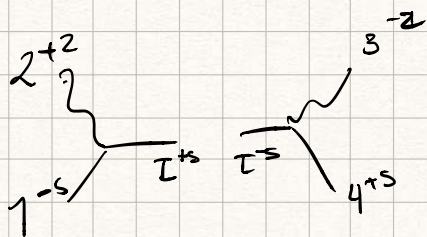
$$A_3 [1^{-5} 2^{-2} 3^{+5}] = g \langle 12 \rangle^{2s+2} \langle 23 \rangle^{2-2s} \langle 31 \rangle^{-2}$$

$$A_3 [1^{+5} 2^{+2} 3^{-5}] = g [12]^{2s+2} [23]^{2-2s} [31]^{-2}$$

note that $P_I = -P_1 - P_2 = P_3 + P_4 \Rightarrow [2|P_I|3] = \begin{cases} -[2|1|3] = +\langle 31 \rangle [12] \\ [2|4|3] = -\langle 34 \rangle [42] \end{cases}$

$$\Rightarrow [2|P_I|3] = \frac{1}{2} [2|(P_4 - P_1)|3]$$

S-channel Residue



$$\begin{aligned} 1) [1+2][2+I][I] &= 0 \\ 3) [3+4][4+I][I] &= 0 \\ 1) [1+2][2+3][3+4][4] &= 0 \end{aligned}$$

$$R_s = A_3 [I^{+5} 2^{+2} 1^{-5}] A_3 [I^{-5} 3^{-2} 4^{+5}]$$

$$= \frac{[I 2]^{2s+2} [2 1]^{2-2s} g^2}{[I 1]^2} \frac{\langle I 3 \rangle^{2s+2} \langle 3 4 \rangle^{2-2s}}{\langle I 4 \rangle^4}$$

$$\begin{aligned} \langle 3 I \rangle [I 2] &= -\langle 31 \rangle [12] \\ &= -\langle 34 \rangle [42] \end{aligned}$$

$$= \frac{(\langle 3 I \rangle [I 2])^{2s+2} (\langle 3 4 \rangle [12])^{2-2s}}{(\langle 4 I \rangle [I 1])^2}$$

$$\begin{aligned} \langle 4 I \rangle [I 1] &= -\langle 4 2 \rangle [21] \\ &= -\langle 4 3 \rangle [31] \end{aligned}$$

$$= \frac{(\langle 13 \rangle [12])^{2s+2} (\langle 34 \rangle [12])^{2-2s}}{(\langle 24 \rangle [12])^{-2}}$$

$$= \frac{[12]^{2s+2+2-2s-2} \langle 13 \rangle^{2s+2} \langle 34 \rangle^{2-2s}}{\langle 24 \rangle^2}$$

$$\langle 34 \rangle [24] = [214|3]$$

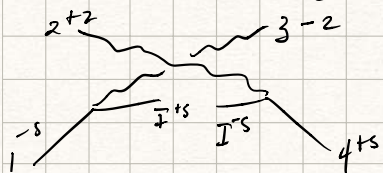
$$= \left(\frac{\langle 13 \rangle [24]}{[24]} \right)^{2s} \frac{[12]^2 \langle 34 \rangle^2 \langle 13 \rangle^2}{\langle 24 \rangle^2 \langle 34 \rangle^{2s}}$$

$$= (\langle 13 \rangle [24])^{2s} \frac{[12]^2 \langle 34 \rangle^2 \langle 13 \rangle^2}{([24] \langle 34 \rangle)^{2s} \langle 24 \rangle^2} \frac{([24] \langle 34 \rangle)^4}{([24] \langle 34 \rangle)^4}$$

$$= (\langle 13 \rangle [24])^{2s} [214|3]^{4-2s}$$

$$= \frac{(\langle 13 \rangle [24])^{2s} [214|3]^{4-2s}}{u^2} \rightarrow - \frac{(\langle 13 \rangle [24])^{2s} [214|3]^{4-2s}}{ut}$$

U-channel Residue



$$\begin{aligned} 1) [1+3][3+I][I] &= 0 \\ 2) [2+4][4+I][I] &= 0 \end{aligned}$$

$$= (s\text{-channel})_{2 \leftrightarrow 3}$$

$$= \frac{(\langle 13 \rangle [24])^{2s} [214|3]^{4-2s}}{s^2}$$

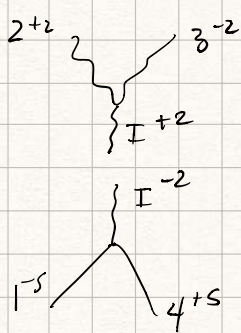
$$= \frac{(\langle 13 \rangle [24])^{2s} \left(\frac{u}{s} \right)^2 [214|3]^{4-2s}}{s^2}$$

the long way:

$$\begin{aligned}
 \frac{R_u}{g^2} &= \frac{A_3 [1^{-5} 3^{-2} I^{+5}]}{g^2} \frac{A_3 [4^{+5} 2^{+2} I^{-5}]}{g^2} \\
 &= \frac{\langle 13 \rangle^{25+2} \langle 3 I \rangle^{2-25}}{\langle I 1 \rangle^2} \frac{[42]^{25+2} [2 I]^{2-25}}{[I 4]^2} \\
 &= \frac{(\langle 3 I \rangle [I 2])^{2-25} (\langle 13 \rangle [24])^{25+2}}{(\langle I 1 \rangle [I 4])^2} \\
 &= (\langle 13 \rangle [24])^{25} \frac{\langle 13 \rangle^2 [24]^2 (\langle 34 \rangle [42])^{2-25}}{\langle 13 \rangle^2 [34]^2} \\
 &= (\langle 13 \rangle [24])^{25} \frac{[24]^2 \langle 34 \rangle^2}{[24]^2 \langle 34 \rangle^2} \\
 &= \frac{(\langle 13 \rangle [24])^{25} [2413]^{4-25}}{g^2} \rightarrow - \frac{(\langle 13 \rangle [24])^{25} [2413]^{4-25}}{g^2}
 \end{aligned}$$

$\langle I 1 \rangle [I 4] = - \langle 13 \rangle [34]$
 $= - \langle 12 \rangle [24]$
 $\langle 3 I \rangle [I 2] = - \langle 31 \rangle [12]$
 $= - \langle 34 \rangle [42]$

the "t"-channel:



$$A_3 [1^{-2} 2^{+2} 3^{-2}] = 2 \langle 12 \rangle^2 \langle 23 \rangle^2 \langle 31 \rangle^6$$

$$A_3 [1^{-5} 2^{-2} 3^{+5}] = g \langle 12 \rangle^{25+2} \langle 23 \rangle^{2-25} \langle 31 \rangle^{-2}$$

$$\begin{aligned}
 1) [1+4][4+I][I=0] \\
 2) [2+3][3+I][I=0]
 \end{aligned}$$

$$\begin{aligned}
 \langle 4 I \rangle [I 3] &= - \langle 41 \rangle [13] \\
 &= - \langle 42 \rangle [23] \\
 \langle 1 I \rangle [I 3] &= - \langle 14 \rangle [43] = - \langle 12 \rangle [23] \\
 \langle 1 I \rangle [I 2] &= - \langle 14 \rangle [42] = - \langle 13 \rangle [32] \\
 \langle 4 I \rangle [I 2] &= - \langle 41 \rangle [12] \\
 &= - \langle 43 \rangle [32]
 \end{aligned}$$

$$\begin{aligned}
 [2|I|3] &= - [2|1|3] = + \langle 31 \rangle [12] \\
 [2|4|3] &= - \langle 34 \rangle [42]
 \end{aligned}$$

$$\begin{aligned}
 \frac{R_t}{g^2} &= \frac{A_3 [1^{-5} I^{-2} 4^{+5}]}{g^2} \frac{A_3 [2^{+2} 3^{-2} I^{+2}]}{g^2} \\
 &= \frac{\langle 1 I \rangle^{25+2} \langle I 4 \rangle^{2-25}}{\langle 4 I \rangle^2} \frac{[I 2]^6}{[23]^2 [3 I]^2} \\
 &= \frac{\langle 1 I \rangle^{25}}{\langle 4 I \rangle^{25}} \frac{\langle 1 I \rangle^2 \langle I 4 \rangle^2 [I 2]^6}{[3 I]^2} \frac{1}{\langle 4 I \rangle^2 [23]^2} \\
 &= \left(\frac{\langle 1 I \rangle [I 2]}{\langle 4 I \rangle [I 2]} \right)^{25} \frac{\langle 1 I \rangle^2 \langle I 4 \rangle^2 [I 2]^6}{[3 I]^2} \frac{1}{\langle 4 I \rangle^2 [23]^2} \\
 &= \frac{\langle 13 \rangle^{25} [32]^{25}}{\langle 43 \rangle^{25} [22]^{25}} \frac{\langle 1 I \rangle^2 \langle I 4 \rangle^2 [I 2]^6}{[3 I]^2} \frac{1}{\langle 4 I \rangle^2 [23]^2} \\
 &= \frac{\langle 13 \rangle^{25} [24]^{25}}{\langle 43 \rangle^{25} [24]^{25}} \frac{\langle 1 I \rangle^2 \langle I 4 \rangle^2 [I 2]^6}{[3 I]^2} \frac{1}{\langle 4 I \rangle^2 [23]^2} \\
 &= (\langle 13 \rangle [24])^{25} [2413]^{-25} \frac{\langle 1 I \rangle^2 \langle I 4 \rangle^2 [I 2]^6}{[3 I]^2} \frac{1}{\langle 4 I \rangle^2 [23]^2} \\
 &= \frac{(\langle 4 I \rangle^2 [I 2]^6) (\langle 1 I \rangle^2 [I 2]^2) [I 2]^2}{\langle 43 \rangle^2 [32]^2 \langle 14 \rangle^2 [42]^2 (\langle 1 I \rangle [I 2])^2} \frac{1}{(\langle 1 I \rangle [I 2])^2}
 \end{aligned}$$



$$\frac{(\langle 43 \rangle [42])^4 (\langle 14 \rangle [23])^2}{(\langle 12 \rangle [12])^2} \frac{[21413]^4 (\langle 14 \rangle [23])^2}{s^2}$$

$$\frac{R_t}{K g} = \frac{(\langle 13 \rangle [24])^{2s}}{s^2} \frac{[21413]^{-2s+4}}{s^2}$$

$$R_s = - \frac{(\langle 13 \rangle [24])^{2s}}{u t} \frac{[21413]^{4-2s}}{s^2} g^2$$

$$R_u = - \frac{(\langle 13 \rangle [24])^{2s}}{s t} \frac{[21413]^{4-2s}}{s^2} g^2$$

$$R_t = - \frac{(\langle 13 \rangle [24])^{2s}}{s u} \frac{[21413]^{4-2s}}{s^2} g^2$$

We can make a consistent ansatz if $g = K$ and

$$A_4 \propto K^2 \frac{(\langle 13 \rangle [24])^{2s}}{s t u} \frac{[21413]^{4-2s}}{s^2}$$

